

QUERY SQL

VERIFICA

- **VERIFICA CONTEGGIO RIGHE**

```
SELECT 'Hotels' AS Tabella, COUNT(*) AS Totale FROM Hotels
UNION ALL
SELECT 'Hotel_Stats', COUNT(*) FROM Hotel_Stats
UNION ALL
SELECT 'Reviewers', COUNT(*) FROM Reviewers
UNION ALL
SELECT 'Reviews', COUNT(*) FROM Reviews;
```

<u>Tabella</u>	<u>Totale</u>
Hotels	1477
Hotel_Stats	1477
Reviewers	16864
Reviews	511944

Le righe corrispondono esattamente a quelle dei file csv da cui sono stati caricati i dati

- **JOIN HOTEL E REVIEWS**

```
SELECT
    H.Hotel_Name,
    R.Review_Date,
    R.Reviewer_Score,
    R.Positive_Review
FROM Reviews AS R
JOIN Hotels AS H ON R.Hotel_ID = H.Hotel_ID
LIMIT 10;
```

<u>Hotel_Name</u>	<u>Review_Date</u>	<u>Reviewer_Score</u>	<u>Positive_Review</u>
Hotel Arena	2017-08-03	2.9	Only the park outside of the hotel was beautiful
Hotel Arena	2017-07-08	7.9	The room was big enough and the bed is good The b...
Hotel Arena	2016-09-19	6.3	beautiful surroundings great location Old world s...
Hotel Arena	2016-09-18	5.0	Location
Hotel Arena	2016-09-12	4.6	Location was ok and has access to all the public ...
Hotel Arena	2016-09-09	8.8	The bed was amazing Actually our whole room large...

<u>Hotel Name</u>	<u>Review Date</u>	<u>Reviewer Score</u>	<u>Positive Review</u>
Hotel Arena	2016-08-31	9.2	super comfy beds cozy rooms lovely staff pretty s...
Hotel Arena	2016-08-13	7.9	Staff were brilliant Really helpful attentive and...
Hotel Arena	2016-07-30	10.0	Great Customer Service Staff very responsive to o...
Hotel Arena	2016-07-27	7.1	The rooms were comfortable and clean and the bed ...

Le Foreign Key risultano collegate bene

- **DISTRIBUZIONE SCORE PER NAZIONALITÀ**

Prendendo in considerazione solo le nazioni con più di 1000 recensioni lasciate e limito la stampa a 5 righe.

SELECT

RW.Reviewer_Nationality,

COUNT(R.Review_ID) AS Totale_Recensioni,

ROUND(AVG(R.Reviewer_Score), 2) AS Media_Voto

FROM Reviews AS R

JOIN Reviewers AS RW ON R.Reviewer_ID = RW.Reviewer_ID

GROUP BY RW.Reviewer_Nationality

HAVING Totale_Recensioni > 100

ORDER BY COUNT(R.Review_ID) DESC

LIMIT 5;

<u>Reviewer_Nationality</u>	<u>Totale_Recensioni</u>	<u>Media_Voto</u>
United Kingdom	244321	8.49
United States of America	35108	8.79
Australia	21502	8.59
Ireland	14733	8.46
United Arab Emirates	10170	7.88

4.3 Analisi con SQL

1. RECENSIONI NEL TEMPO

- **ANDAMENTO RECENSIONI PER MESE E ANNO**

SELECT

Year,

Month,

```

COUNT(Review_ID) AS Numero_Recensioni
FROM Reviews
GROUP BY Year, Month
ORDER BY Year DESC, Month DESC;

```

- HOTEL CON PICCO RECENSIONI

Hotel_Name	Year	Month	PICCO_RECENSIONI	1
Strand Palace Hotel	2016	2	304	
Intercontinental London The O2	2016	4	278	
Strand Palace Hotel	2016	3	267	
Britannia International Hotel Canary Wharf	2015	11	258	
Park Plaza Westminster Bridge London	2016	8	257	
Britannia International Hotel Canary Wharf	2017	5	248	
Strand Palace Hotel	2016	1	242	
Britannia International Hotel Canary Wharf	2017	3	241	
Britannia International Hotel Canary Wharf	2016	5	239	
Britannia International Hotel Canary Wharf	2017	6	239	

2. NAZIONALITÀ RECENSORI

- TOP 10 NAZIONALITÀ PER NUMERO RECENSIONI E SCORE MEDIO

```

SELECT RR.Reviewer_Nationality AS 'TOP NAZIONALITA' RECENSORI', COUNT(R.Review_ID) AS 'TOTALE RECENSIONI',
ROUND(AVG(R.Reviewer_Score ),2) AS 'VOTO MEDIO'
FROM Reviewers AS RR
JOIN Reviews AS R ON RR.Reviewer_ID = R.Reviewer_ID
GROUP BY RR.Reviewer_Nationality
ORDER BY COUNT(R.Review_ID)DESC
LIMIT 10;

```

TOP NAZIONALITA' RECENSORI	TOTALE RECENSIONI	VOTO MEDIO
UNITED KINGDOM	244321	8.49
UNITED STATES OF AMERICA	35108	8.79
AUSTRALIA	21502	8.59
IRELAND	14733	8.46
UNITED ARAB EMIRATES	10170	7.88
SAUDI ARABIA	8903	7.88

<u>TOP NAZIONALITA' RECENSORI</u>	<u>TOTALE RECENSIONI</u>	<u>VOTO MEDIO</u>
NETHERLANDS	8691	8.13
SWITZERLAND	8607	8.16
GERMANY	7831	8.13
CANADA	7802	8.55

- % CONTRIBUTO TOP NAZIONALITÀ AL TOTALE RECENSIONI**

```

SELECT
  RR.Reviewer_Nationality AS 'NAZIONALITÀ',
  COUNT(R.Review_ID) AS 'RECENSIONI',
  ROUND(
    (COUNT(R.Review_ID) * 100.0 / SUM(COUNT(R.Review_ID)) OVER()),
    2
  ) AS '% CONTRIBUTO'
FROM Reviewers AS RR
JOIN Reviews AS R ON RR.Reviewer_ID = R.Reviewer_ID
GROUP BY RR.Reviewer_Nationality
ORDER BY COUNT(R.Review_ID) DESC
LIMIT 10;

```

<u>NAZIONALITÀ</u>	<u>RECENSIONI</u>	<u>% CONTRIBUTO</u>
United Kingdom	244321	47.72
United States of America	35108	6.86
Australia	21502	4.20
Ireland	14733	2.88
United Arab Emirates	10170	1.99
Saudi Arabia	8903	1.74
Netherlands	8691	1.70
Switzerland	8607	1.68
Germany	7831	1.53
Canada	7802	1.52

- CONFRONTO SCORE MEDIO UK VS ALTRI**

```

SELECT
  CASE
    WHEN RR.Reviewer_Nationality = ' United Kingdom ' THEN 'United Kingdom'
    ELSE 'Rest of the World'
  END AS Gruppo_Nazionalita,
  COUNT(R.Review_ID) AS Totale_Recensioni,
  ROUND(AVG(R.Reviewer_Score), 2) AS Score_Medio,
  ROUND(
    (COUNT(R.Review_ID) * 100.0 / SUM(COUNT(R.Review_ID)) OVER()),
    2
  ) AS Percentuale_Contributo
FROM Reviewers AS RR
JOIN Reviews AS R ON RR.Reviewer_ID = R.Reviewer_ID

```

GROUP BY Gruppo_Nazionalita
ORDER BY Totale_Recensioni DESC;

<u>Gruppo Nazionalita</u>	<u>Totale Recensioni 1</u>	<u>Score Medio</u>	<u>Percentuale Contributo</u>
Rest of the World	267623	8.31	52.28
United Kingdom	244321	8.49	47.72

3. HOTEL E PERFORMANCE

- HOTEL CON MAGGIOR AVERAGE_SCORE E NUMERO RECENSIONI

```
SELECT h.Hotel_Name, hs.Total_Number_of_Reviews, hs.Average_Score
FROM Hotels AS h
JOIN Hotel_Stats AS hs ON h.Hotel_ID = hs.Hotel_ID
ORDER BY hs.Average_Score DESC, hs.Total_Number_of_Reviews DESC
LIMIT 5;
```

<u>HOTEL_NAME</u>	<u>TOTAL NUMBER OF REVIEWS</u>	<u>AVERAGE SCORE 1</u>
RITZ PARIS	122	9.8
HOTEL CASA CAMPER	732	9.6
HOTEL THE SERRAS	604	9.6
H10 CASA MIMOSA 4 SUP	454	9.6
HAYMARKET HOTEL	255	9.6

- ANALISI GAP TRA REVIEWER SCORE E AVERAGE SCORE

```
SELECT
  H.Hotel_Name,
  ROUND(AVG(HS.Average_Score), 2) AS Score_Ufficiale,
  ROUND(AVG(R.Reviewer_Score), 2) AS Media_Recensioni,
  ROUND(AVG(R.Reviewer_Score) - AVG(HS.Average_Score), 2) AS GAP
FROM Reviews AS R
JOIN Hotels AS H ON R.Hotel_ID = H.Hotel_ID
JOIN Hotel_Stats AS HS ON H.Hotel_ID = HS.Hotel_ID
GROUP BY H.Hotel_ID
HAVING COUNT(R.Review_ID) > 50 – prendiamo in considerazione solo hotel che hanno almeno 50 recensioni
ORDER BY GAP ASC – mette in risalto chi ha deluso di più
LIMIT 10;
```

<u>Hotel_Name</u>	<u>Score_Ufficiale</u>	<u>Media_Recensioni</u>	<u>GAP 1</u>
Mercure Paris 19 Philharmonie La Villette	7.50	6.91	-0.59
Meli Milano	8.30	7.72	-0.58
Hotel Parco di Sch nbrunn Vienna	7.90	7.35	-0.55
Hotel Tiziano Park Vita Parcour Gruppo MiniHotels	8.60	8.07	-0.53
Hotel OFF Paris Seine	8.40	7.88	-0.52

Hotel Name	Score Ufficiale	Media Recensioni	GAP 1
Klima Hotel Milano Fiere	8.30	7.79	-0.51
Hotel Vienna	8.10	7.61	-0.49
Mercure Paris Tour Eiffel Pont Mirabeau	8.00	7.52	-0.48
Gran Hotel Barcino	7.00	6.52	-0.48
Hotel Sunflower	7.60	7.14	-0.46

4. GEOGRAFIA

- DISTRIBUZIONE RECENSIONI PER CITTÀ

```
SELECT h.Hotel_City AS Città,
COUNT(r.Review_ID) AS Numero_Recensioni,
ROUND(COUNT(r.Review_ID) * 100.0 / (SELECT COUNT(*) FROM Reviews),
2) AS Percentuale_totale_recensioni
FROM Hotels AS h
JOIN Reviews AS r ON h.Hotel_ID = r.Hotel_ID
GROUP BY h.Hotel_City
ORDER BY COUNT(r.Review_ID) DESC;
```

Città	Numero_Recensioni	Percentuale_totale_recensioni
London	262298	51.24
Barcelona	59942	11.71
Paris	59031	11.53
Amsterdam	57211	11.18
Milan	37204	7.27
Vienna	36258	7.08

- SCORE MEDIO PER NAZIONE, TOP CITTÀ

```
SELECT
h.Hotel_Country AS Nazione,
h.Hotel_City AS Città,
COUNT(r.Review_ID) AS Numero_Recensioni,
ROUND(AVG(hs.Average_Score), 2) AS Score_medio
FROM Hotels AS h
JOIN Hotel_Stats AS hs ON h.Hotel_ID = hs.Hotel_ID
JOIN Reviews AS r ON h.Hotel_ID = r.Hotel_ID
GROUP BY h.Hotel_Country, h.Hotel_City
ORDER BY Score_medio DESC;
```

Nazione	Città	Numero_Recensioni	Score_medio 1
Austria	Vienna	36258	8.58

Spain	Barcelona	59942	8.52
Italy	Milan	37204	8.43
France	Paris	59031	8.41
Netherlands	Amsterdam	57211	8.39
United Kingdom	London	262298	8.34

5. LUNGHEZZA RECENSIONI

• DISTRIBUZIONE REVIEW_LENGTH PER CATEGORIA SCORE

```

SELECT
  hs.Average_Score AS Categoria_Score,
  ROUND(AVG(r.Review_Total_Negative_Word_Counts + r.Review_Total_Positive_Word_Counts), 0) AS Lunghezza_Media_Recensione,
  COUNT(r.Review_ID) AS Numero_Recensioni
FROM Reviews AS r
JOIN Hotel_Stats AS hs ON r.Hotel_ID = hs.Hotel_ID
GROUP BY hs.Average_Score
ORDER BY hs.Average_Score DESC;

```

<u>Categoria Score</u>	<u>Lunghezza Media Recensione</u>	<u>Numero Recensioni</u>
9.8	33	28
9.6	37	915
9.5	33	1207
9.4	35	9339
9.3	35	12659
9.2	34	12935
9.1	36	21379
9.0	35	20960
8.9	36	28324
8.8	37	30546
8.7	36	37798
8.6	35	36320
8.5	35	37821
8.4	36	41164
8.3	37	31945
8.2	36	34847
8.1	37	38070
8.0	37	21878

<u>Categoria_Score</u>	<u>Lunghezza_Media_Recensione</u>	<u>Numero_Recensioni</u>
7.9	37	14848
7.8	37	19468
7.7	39	11829
7.6	37	6651
7.5	38	13719
7.4	37	6300
7.3	40	3997

• CORRELAZIONE LUNGHEZZA RECENSIONE VS REVIEWER_SCORE

```

SELECT
  ROUND(r.Reviewer_Score, 0) AS Fascia_Voto,
  ROUND(AVG(r.Review_Total_Negative_Word_Counts + r.Review_Total_Positive_Word_Counts), 0) AS Lunghezza_Media_Parole,
  COUNT(*) AS Numero_Recensioni
FROM Reviews AS r
GROUP BY Fascia_Voto
ORDER BY Fascia_Voto DESC;

```

<u>Fascia_Voto_1</u>	<u>Lunghezza_Media_Parole</u>	<u>Numero_Recensioni</u>
10	31	186594
9	35	105382
8	36	108716
7	42	44101
6	48	27244
5	51	24122
4	56	9236
3	57	6549